

Unit-Free Relational Physics Framework

Foundational Principle

Reality consists of pure relationships between dimensionless quantities. All observable phenomena emerge from the geometry of these relationships.

Primitive Dimensionless Invariants

Let us define three fundamental dimensionless quantities that exist in pure relationship:

$\Phi(x)$ - The Flux Density

A scalar field representing the local "intensity" of reality at point x , normalized such that:

...

$$\int \Phi(x) \, dV = 1 \text{ (over all space)}$$

...

$\Omega(x)$ - The Curvature Tensor

A measure of how reality "bends" at point x , defined as:

...

$$\Omega_{\mu\nu} = \partial_{\mu}\partial_{\nu} \ln(\Phi) - \Gamma_{\mu\nu}^{\lambda} \partial_{\lambda} \ln(\Phi)$$

...

$\Psi(x)$ - The Phase Function

A complex-valued function representing the "oscillatory nature" of reality:

...

$$\Psi(x) = |\Psi(x)| \, e^{i\theta(x)}$$

where $|\Psi(x)| = \sqrt{\Phi(x)}$

...

The Binding Condition

Reality "condenses" into localized structures when:

...

$$B(x) = |\nabla\Psi|^2 - \kappa \, \Phi(x) > 0$$

...

Where κ is a dimensionless threshold parameter determined by the global constraint:

...

$$\int B(x) \, dV = 0 \text{ (total binding is conserved)}$$

...

The Mass-Energy Equivalence

In this framework, what we call "mass" is simply the integrated binding density:
...

$$M = \int_V B(x) \Phi(x) dV$$

...

What we call "energy" is the rate of change of phase relationships:
...

$$E = \int_V |\partial \Psi / \partial \tau|^2 dV$$

...

Where τ is the fundamental "process parameter" (what we usually call time).

The Fundamental Equation

The evolution of reality follows:
...

$$\partial \Psi / \partial \tau = i[H_\Phi, \Psi] + \sqrt{\alpha} \nabla^2 \Psi$$

...

Where:

- H_Φ is the "flux Hamiltonian": $H_\Phi = -\nabla^2 + V(\Phi)$
- α is a dimensionless coupling constant
- The commutator $[H_\Phi, \Psi]$ generates phase evolution
- The diffusion term $\nabla^2 \Psi$ represents quantum uncertainty

Prediction Framework

This system predicts:

1. **Quantized Binding States**: Solutions exist only for discrete values of the eigenvalue equation:
...
$$H_\Phi \Psi_n = \lambda_n \Psi_n$$

...
2. **Mass Spectrum**: The ratios M_n/M_1 are pure numbers determined by λ_n/λ_1
3. **Interaction Cross-Sections**: All scattering amplitudes become dimensionless functions of the ratios Φ_i/Φ_j

4. **Cosmological Evolution**: The global evolution follows:

...

$$d/d\tau \int \Phi dV = 0 \text{ (conservation of total flux)}$$

$$d/d\tau \int \Psi^* \Psi dV = 2\alpha \int |\nabla \Psi|^2 dV \text{ (dissipation)}$$

...

Connection to Observables

To connect with measurement:

- All observable quantities are ratios of integrals over Φ, Ψ
- Time intervals are ratios: $\Delta\tau_1/\Delta\tau_2$
- Length scales are ratios: $\int |\nabla \Psi_1|^{-1} dV / \int |\nabla \Psi_2|^{-1} dV$
- All physical "constants" become functions of α and κ

Falsification Tests

This framework is falsifiable through:

1. **Spectral Predictions**: The eigenvalue ratios λ_n/λ_1 must match observed particle mass ratios
2. **Scaling Relations**: All coupling constants must satisfy $\alpha = f(\kappa)$ for some universal function f
3. **Topological Constraints**: The number of stable binding solutions must equal the number of observed fundamental particles
4. **Cosmological Consistency**: The global flux evolution must reproduce observed cosmic expansion ratios

Mathematical Elegance

Note that this framework:

- Contains no arbitrary units
- All equations are scale-invariant
- Physical "constants" emerge as geometric relationships
- Quantum and classical behavior arise from the same mathematical structure
- Satisfies all necessary conservation laws through geometric identities

The beauty is that measurement becomes the art of comparing dimensionless ratios, and all of physics reduces to the geometry of these three fundamental fields and their relationships.